

Lecture 15 - EM Algorithm & Factor analysis

Outline

- EM convergence
- Factor analysis
- EM steps

EM Algorithm recap.

E-step

$$Q_i(z^{(i)}) = P(z^{(i)} | x^{(i)}; \theta)$$

M-step

$$\theta_1 = \arg \max_{\theta} \sum_i \sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})}$$

$$P(x^{(i)}, z^{(i)}) = P(x^{(i)} | z^{(i)}) \cdot P(z^{(i)})$$

$$z^{(i)} \sim \text{Multinomial}(\phi)$$

$$[P(z^{(i)} = j) = \phi_j]$$

$$x^{(i)} | z^{(i)} = j \sim N(\mu_j, \Sigma_j)$$

$$\text{E-step} \Rightarrow Q_i(z^{(i)} = j) = P(z^{(i)} = j | x^{(i)}; \phi, \mu, \Sigma)$$

$$P(z^{(i)} = j)$$

M-step

$$\max_{\phi, \mu, \Sigma} \sum_i \sum_{z^{(i)}} \phi_i(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \phi, \mu, \Sigma)}{\phi_i(z^{(i)})}$$

$$= \sum_i \sum_j w_j^{(i)} \log \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x^{(i)} - \mu_j)^T \Sigma_j^{-1} (x^{(i)} - \mu_j)\right) \phi_j$$

$w_j^{(i)}$

where $p(x^{(i)} | z^{(i)}; \phi, \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x^{(i)} - \mu_j)^T \Sigma_j^{-1} (x^{(i)} - \mu_j)\right) \phi_j$

and $\phi_j = p(z^{(i)})$

Take derivative of above & set it to 0.

$$\nabla_{\mu_j} (\dots) = 0$$

$$\mu_j = \frac{\sum_i w_j^{(i)} x^{(i)}}{\sum_i w_j^{(i)}}$$

$w_j^{(i)}$ = "strength with which $x^{(i)}$ is assigned to Gaussian j "

$$p(z^{(i)} = j | x^{(i)}; \dots)$$

$$\nabla_{\phi} (\dots) = 0$$

$$\phi_j = \frac{\sum_i w_j^{(i)}}{\sum_i \sum_j w_j^{(i)}}$$

In factor analysis model ~~is~~, $z^{(i)}$ is continuous, and it will be a gaussian distribution.

In the m -step, for the value of θ , instead of summation, it would be an integral.

By Jensen's inequality we can prove that

$$J(\theta, Q) = \sum_i \sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})}$$

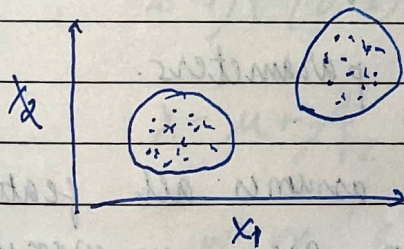
We know $\ell(\theta) \geq J(\theta, Q)$ holds true for any θ, Q

Applying the E-M algorithm,

in the E-step: maximize J wrt Q_i

in the m-step: maximize J wrt θ

If $m \gg n$, mixture of Gaussians can be used.



but $m \approx n$ or $m \ll n$

Say $m = 30$, $n = 100$.

Model as single Gaussian.

$$x \sim N(\mu, \Sigma)$$

$$\text{MLE: } \mu = \frac{1}{M} \sum_i x^{(i)}$$

$$\Sigma = \frac{1}{m} (x^{(i)} - \mu)(x^{(i)} - \mu)^T$$

If $m \leq n$, the Σ will be singular/non-invertible.

hence, MLE won't work.

So as an alternate $\rightarrow$ Option 1: consider Σ to be diagonal

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & 0 & \sigma_n^2 \end{bmatrix}$$

$$\text{MLE: } \sigma_j^2 = \frac{1}{m} \sum_i (x_j^{(i)} - \mu_j)^2$$

It has n parameters.

But this model assumes all features such as $n_1, n_2, \dots, n_n$ are un-correlated, which is not a great assumption.

Option d: constrain Σ to be $\Sigma = \sigma^2 I$

$$\text{i.e. } \Rightarrow \Sigma = \begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{bmatrix}$$

Across all its diagonals, it has only 1 entry.

$$\text{MSE} = \sigma^2 = \frac{1}{mn} \sum_i \sum_j (x_{ij} - \mu_j)^2$$

This model is a much stronger assumption than model 1, hence it's a much worse model.

Factor Analysis

$$P(x, z) = P(x|z) \cdot P(z)$$

z is hidden

$d=3$
 $n=100$
 $m=30$

$$z \sim \mathcal{N}(0, I) \quad z \in \mathbb{R}^d \quad (d < n)$$

$$x = \mu + A z + \epsilon$$

where $\epsilon \sim \mathcal{N}(0, \Psi)$

Parameters

$$\mu \in \mathbb{R}^n, A \in \mathbb{R}^{n \times d}, \Psi \in \mathbb{R}^{n \times n}$$

diagonal

Equivalently, $x|z \sim \mathcal{N}(\mu + \Lambda_z, \Sigma)$

Examples

$z \in \mathbb{R}^1, x \in \mathbb{R}^2, d=1, n=d, m=7$

Say $\Lambda = \begin{bmatrix} \alpha \\ 1 \end{bmatrix}, \mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Lambda_z + \mu \in \mathbb{R}^2$

Multivariate Gaussian

$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \begin{matrix} \updownarrow \pi \\ \updownarrow s \end{matrix} \quad \begin{matrix} x_1 \in \mathbb{R}^\pi, x_2 \in \mathbb{R}^s \\ x \in \mathbb{R}^{\pi+s} \end{matrix}$

$x \sim \mathcal{N}(\mu, \Sigma)$

$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \begin{matrix} \updownarrow \pi \\ \updownarrow s \end{matrix}$

$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{matrix} \updownarrow \pi \\ \updownarrow s \end{matrix}$

Marginal: $P(x_1) = ?$

$P(x) = P(x_1, x_2)$

$\int_{x_2} P(x_1, x_2) dx_2 = P(x_1)$

$\frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} \begin{pmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{pmatrix}^T \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{pmatrix} \right\}$

$$P(x_1)$$

$$x_1 \sim \mathcal{N}(\mu_1, \Sigma_1)$$

$$\text{conditional: } P(x_1 | x_2) = ?$$

$$\frac{P(x_1, x_2)}{P(x_2)}$$

$$x_1 | x_2 \sim \mathcal{N}(\mu_{1|2}, \Sigma_{1|2})$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

1) Derive $P(x, z)$

$$\begin{pmatrix} x \\ z \end{pmatrix} \sim \mathcal{N}(\mu_{x,z}, \Sigma)$$

$$z \sim \mathcal{N}(0, I) \quad E_z = 0$$

$$x = \mu + \Lambda_z + \Sigma \quad E_x = E\left[\mu + \Lambda \frac{1}{z} + \Sigma\right]$$

$$E_x = E[\mu] = \mu$$

$$\mu_{xz} = \begin{bmatrix} 0 \\ \mu \end{bmatrix} \begin{matrix} \uparrow d \\ \uparrow n \end{matrix}$$

$$E(z - E_z)(x - E_x)^T$$

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} = \begin{bmatrix} E(z - E_z)(z - E_z)^T & \\ E(x - E_x)(z - E_z)^T & \\ E(x - E_x)(x - E_x)^T & \end{bmatrix}$$

$$E(x - E_x)(x - E_x)^T$$

$$= \begin{bmatrix} I & \Lambda^T \\ \Lambda & \Lambda \Lambda^T + \Psi \end{bmatrix}$$

$$\begin{bmatrix} z \\ x \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ \mu \end{bmatrix}, \begin{bmatrix} I & \Lambda^T \\ \Lambda & \Lambda \Lambda^T + \Psi \end{bmatrix} \right)$$

E-Step:

$$Q_i(z^{(i)}) = p(z^{(i)} | x^{(i)}; \theta)$$

$$z^{(i)} | x^{(i)} \sim \mathcal{N} \left(\mu_{z^{(i)} | x^{(i)}}, \Sigma_{z^{(i)} | x^{(i)}} \right)$$

where

$$\mu_{z^{(i)} | x^{(i)}} = \vec{0} + \Lambda^T (\Lambda \Lambda^T + \Psi)^{-1} (x^{(i)} - \mu)$$

$$\Sigma_{z^{(i)} | x^{(i)}} = I - \Lambda^T (\Lambda \Lambda^T + \Psi)^{-1} \Lambda$$

$$M\text{-step: } \mu = \mathbb{E}[z] = x$$